



The Exploit Framework for Penetration Testers

Today, the Metasploit Framework is considered the single most useful auditing tool that is freely available to security professionals and penetration testers. It has a wide array of commercial-grade exploits, an extensive exploit-development environment with tools to gather network information, and Web vulnerability plug-ins. This article is an introduction to the framework.

The Metasploit Framework (MSF) provides the ability to launch exploits against selected target systems, and to perform post-exploitation tasks, such as uploading files, running processes, establishing back-door network connections, monitoring system use, and many more. Therefore, its primary use is in the penetration testing process.

Another important use of the MSF is in systems administration. So far, the development of exploits has been limited to a select group of people within the security research, hacking and testing communities. With the help of a reliable exploitation platform like Metasploit, administrators are now able to check multiple servers for vulnerability to a given exploit, and what's more, they can even go to the extent of running the exploit, to determine if the systems are indeed vulnerable.

User interfaces

There are many different interfaces available to the MSF, each with their own strengths and weaknesses. As such, there is no one perfect interface to use with MSF, although the *msfconsole* is considered to be the only supported way to access most features of the MSF.

Msfconsole

The *msfconsole* (see Figure 1) is probably the most popular interface to the MSF. It provides an 'all-in-one' centralised console. It is the traditional and primary means of using the MSF, and is the only supported way to access most of the features of Metasploit. It is the most stable MSF Interface.

After installation, launch it by running *./msfconsole* (from within the directory where it has been installed).

Msfgui

Msfgui is, as the name implies, the graphical user interface of the framework. It is a good tool for demonstrations to clients and management; it provides a point-and-click interface for exploitation, and a GTK wizard-based interface to use the MSF.

Msfweb

The *msfweb* interface (see Figure 2) is the only GUI currently available for the MSF version after 3.3. It offers no security whatsoever, but is currently the recommended way to use the framework on Windows. This interface can be launched with a number of options. You can launch *msfweb* with the *./msfweb* command in a *NIX environment. This UI supports multiple users, and has an AJAX-based *msfconsole* implementation.

Msfcli

Msfcli provides a powerful command-line interface to the framework. It supports the launching of exploits and auxiliary modules. It is excellent if you know exactly which exploit and option you need, and wonderful for use in scripts and basic automation. Note that when using *msfcli*, variables are case-sensitive, and are assigned using '='.

Commands in msfconsole

show

Entering *show* at the *msfconsole* prompt will display every module within Metasploit: